

Predictive Insight Engine – Theory Part

1. What are Supervised Learning Algorithms?

Supervised Learning Algorithms are machine learning algorithms that learn from labeled data. These algorithms are trained using input data and corresponding output labels. After training, the model can predict outputs for new unseen data.

Supervised learning is mainly divided into:

- Regression
- Classification

Examples:

- House price prediction
- Spam email detection
- Weather forecasting

2. Difference Between Regression and Classification

Regression	Classification
Predicts continuous values	Predicts categorical values
Output is numeric	Output is labels/classes
Example: House price prediction	Example: Spam detection
Uses algorithms like Linear Regression	Uses algorithms like Logistic Regression

3. Explain Simple Linear Regression

Simple Linear Regression is a supervised learning algorithm used to predict a continuous dependent variable using one independent variable.

It finds the best-fit straight line between input and output variables.

Formula

$$y = mx + c$$

Where:

- y = predicted value
 - x = independent variable
 - m = slope of line
 - c = intercept
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4. Assumptions of Linear Regression

1. Linear relationship between variables
 2. Independence of errors
 3. Homoscedasticity
 4. No multicollinearity
 5. Errors should be normally distributed
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5. Bias–Variance Trade-Off

Bias–Variance Trade-Off is the balance between underfitting and overfitting.

High Bias

- Model is too simple
- Causes underfitting

High Variance

- Model is too complex
- Causes overfitting

A good machine learning model balances both.

6. Overfitting and Underfitting

Underfitting

Occurs when the model is too simple and cannot capture data patterns.

Overfitting

Occurs when the model memorizes training data including noise.

7. Independent and Dependent Variables

Independent Variables

- Area
- Bedrooms
- Bathrooms
- Location Score
- Property Age

Dependent Variable

- House Price
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8. Data Visualization

Data visualization helps in understanding patterns and relationships.

Common Techniques

- Scatter Plot
 - Heatmap
 - Histogram
 - Pair Plot
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9. Train-Test Split

The dataset is divided into:

- Training Data
- Testing Data

Usually:

- 80% Training
 - 20% Testing
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10. Model Evaluation Metrics

Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Lower MSE indicates better performance.

Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Lower MAE indicates fewer prediction errors.

Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Lower RMSE indicates better accuracy.

R² Score

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Higher R² indicates better model performance.

Adjusted R² Score

$$Adjusted R^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}$$

Where:

- n = number of observations
 - p = number of features
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11. Multiple Linear Regression

Multiple Linear Regression predicts output using multiple independent variables.

Formula

$$y = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$$

12. Polynomial Regression

Polynomial Regression is used when the relationship between variables is nonlinear.

Formula

$$y = a + bx + cx^2 + dx^3 + \dots$$

13. Gradient Descent

Gradient Descent is an optimization algorithm used to minimize the cost function.

Formula

$$\theta = \theta - \alpha \frac{\partial J(\theta)}{\partial \theta}$$

Where:

- θ = parameter
 - α = learning rate
 - $J(\theta)$ = cost function
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14. Types of Gradient Descent

Batch Gradient Descent

Uses the complete dataset.

Stochastic Gradient Descent (SGD)

Uses one training example at a time.

Mini-Batch Gradient Descent

Uses small batches of data.

15. Comparison of Gradient Descent Methods

Method	Speed	Stability
Batch GD	Slow	Stable
SGD	Fast	Noisy
Mini-Batch GD	Moderate	Balanced

16. Bias and Variance Analysis

Model	Bias	Variance
Simple Linear Regression	High	Low
Multiple Linear Regression	Moderate	Moderate
Polynomial Regression	Low	High

17. Best Model for This Project

Multiple Linear Regression performed best because it:

- Uses multiple features
 - Reduces prediction error
 - Balances bias and variance
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18. Final Conclusion

This project successfully implemented:

- Simple Linear Regression
- Multiple Linear Regression
- Polynomial Regression
- Gradient Descent Optimization
- Model Evaluation Metrics
- Bias-Variance Analysis

Among all models, Multiple Linear Regression provided the best performance because it used multiple features and achieved higher accuracy.

The project demonstrates how machine learning can be used for accurate house price prediction.